Tier 4: Model Registry & Cached Datasets

**JSON Artifacts Bundle (artifacts/)**
model.json · preprocessor.json · thresholds.json · model_card.json

**Evaluation Cache (predictions/ & data/)**
demo_sample.parquet (41k rows) · test_predictions.parquet (954k rows)

Tier 1: Operational Workspace & UI Layer (Streamlit)

Transaction Scoring Workspaces

**2. Table Builder**
Editable multi-row sandbox grid

**1. Single Scorer**
Interactive form & real test profiles

**3. Batch Upload**
CSV switch ingestion & matrix check

Governance & Policy Workspaces

**7. Model Card**
Provenance, metadata & limitations

**6. Experiment Gallery**
31 figures & 41 notebook tables

**5. Test Set Explorer**
4-quadrant confusion matrix audit

**4. Threshold & Risk**
Live R-slider over 954k test rows

Tier 2: Core Brokerage & Guard Layer (prototype/core/)

**ui.py Renderer**
Verdict pills, diverging bars & alert banners

**scorer.py Singleton**
@st.cache_resource · Zero session drift

**schema.py Contract**
7 switch fields · Drops leaky origin balances

**explain.py Engine**
Translates SHAP log-odds to plain English

Tier 3: Inference & Serving Engine (src/momo_fraud/)

**XGBoost Classifier**
300 trees · depth 6 · unweighted training

**FeatureBuilder**
transform_one() ≡ batch transform
14 ablated features · Zero-skew guarantee

**FraudScorer API**
predict_proba() · Risk score S = 100·p

**TreeSHAP Explainer**
Local attribution: $f(x) = \varphi_0 + \Sigma\varphi_i$

Operational Authentication Escalation Ladder

**Critical Band (S > 1.29)**
High fraud density · Quarantine Block

**Low Band (S ≤ 0.0129)**
97.04% traffic · Automatic Allow

**High Band (0.129 < S ≤ 1.29)**
1.39% traffic · Step-Up Auth Challenge

**Medium Band (0.0129 < S ≤ 0.129)**
1.57% traffic · Silent Monitoring